To what extent is the rate of adoption of AI at the workplace associated with the increase in the unemployment rate? A comparison of youth and the total labour force

Tutorial 6 Group 2
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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Artificial Intelligence (AI) is changing the way how firms hire workers, manage and organise work and use technology which has led to an increasingly important social and economic issue. With recent developments in generative AI, especially after the release of ChatGPT, in Novemeber of 2022, AI tools are more visible and widely used across different industries. It is argued that AI in the workplace increases productivity but it also may reduce demand of some type of labour, especially in entry level jobs (Cazzaniga et al ., 2024).

This becomes increasingly relevant for young workers aged 18-24. As workers in this age group, often have low to nil level of work experience and are very likely to work in temporary, entry level jobs. Therefore, this results in these workers being vulnerable when employers change the skills that they demand or when certain tasks can be fulfilled by AI. Youth Employment is therefore a social problem, because unemployment in early stages of their career/life can negatively affect future earnings, skill development and work experience which is required for higher skilled jobs (International Labour Organisation, 2024).

This project examines whether the rise of national AI-Development capacity is linked and associated with worse labour outcomes for young workers compared to the total labour force. We compare data from countries from the years 2021-2025. The aim is whether there is an observable association between AI adoption and youth unemployment relative to overall unemployment.

Part 2 - Data Sourcing

2.1 Description and Limitations

The data was compiled from the following datasets:

The World Bank Data:
- Unemployment, youth total (% of total labor force ages 15-24) (modeled ILO estimate): `SL.UEM.1524.ZS`
- Unemployment, total (% of total labor force) (modeled ILO estimate): `SL.UEM.TOTL.ZS`

Eurostat:
- AI adoption rate, or AI used in enterprises: `isoc_eb_ai`

Table 1: Datasets used in this report

Dataset	Unit of observation	Key variables	Source	Period
World Bank (WDI)	Country-year	total_unemp, youth_unemp (% of labour force)	World Bank / ILO modelled estimates (ILOSTAT)	1991-2024 (annual)
Eurostat (isoc_eb_ai)	Country-year	values (% of enterprises using AI)	Eurostat: ICT usage in enterprises survey	2021, 2023, 2024, 2025

Limitations The Eurostat dataset of AI adoption rate only consists of European countries with temporal data from 2021 to 2025. Because of this, the findings cannot be generalised to other regions and the whole world. A wider dataset with more temporal data points can help increase accuracy of the findings and produce more predictive outcomes, but since the wave of AI usage is a fairly new societal shift, it is reasonable that there are currently few temporal data to observe. Another point worth mentioning is that AI adoption rate in 2022 is unavailable from the source, so we lack one important temporal value of the year ChatGPT was launched. To accommodate this, this paper looks at any difference of the association of AI usage and unemployment rate before and after the launch of ChatGPT, which works as the main event that signals the rise of the AI wave.

2.2 Loading the data

```
## Rows: 1,156
## Columns: 14
## $ country     <chr> "Albania", "Albania", "Albania", "Albania", "Albania", "Al~
## $ iso2c       <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL"~
## $ iso3c       <chr> "ALB", "ALB", "ALB", "ALB", "ALB", "ALB", "ALB", "ALB", "A~
## $ year        <int> 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000~
## $ status      <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", ""~
## $ lastupdated <chr> "2026-04-08", "2026-04-08", "2026-04-08", "2026-04-08", "2~
## $ total_unemp <dbl> 10.304, 30.007, 25.251, 20.835, 14.607, 13.928, 16.872, 20~
## $ youth_unemp <dbl> 15.687, 45.621, 38.820, 32.402, 23.005, 22.071, 26.610, 31~
## $ region      <chr> "Europe & Central Asia", "Europe & Central Asia", "Europe ~
## $ capital     <chr> "Tirane", "Tirane", "Tirane", "Tirane", "Tirane", "Tirane"~
## $ longitude   <chr> "19.8172", "19.8172", "19.8172", "19.8172", "19.8172", "19~
## $ latitude    <chr> "41.3317", "41.3317", "41.3317", "41.3317", "41.3317", "41~
## $ income      <chr> "Upper middle income", "Upper middle income", "Upper middl~
## $ lending     <chr> "IBRD", "IBRD", "IBRD", "IBRD", "IBRD", "IBRD", "IBRD", "I~
```

```
## Rows: 72,527
## Columns: 8
## $ freq        <chr> "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A"~
## $ size_emp    <chr> "0-9", "0-9", "0-9", "0-9", "0-9", "0-9", "0-9", "0-9", "0~
## $ nace_r2     <chr> "C10-S951_X_K", "C10-S951_X_K", "C10-S951_X_K", "C10-S951_~
## $ indic_is    <chr> "E_AIX_CC1SIX_DA", "E_AIX_CC1SIX_DA", "E_AIX_CC1SIX_DA", "~
## $ unit        <chr> "PC_ENT", "PC_ENT", "PC_ENT", "PC_ENT", "PC_ENT_IUSE", "PC~
## $ geo         <chr> "ES", "ES", "ES", "PT", "ES", "ES", "ES", "PT", "ES", "ES"~
## $ TIME_PERIOD <dbl> 2023, 2024, 2025, 2023, 2023, 2024, 2025, 2023, 2023, 2024~
## $ values      <dbl> 5.74, 5.66, 5.49, 13.36, 6.88, 6.94, 6.97, 15.23, 1.40, 1.~
```

Table 2: Size of each raw dataset

Dataset	Rows	Columns
wb_raw (World Bank)	1156	14
ai_raw (Eurostat)	72527	8

2.3 Describe the type of variables included

Unemployment rate - World Bank
Two indicators are used in this report: total unemployment (`SL.UEM.TOTL.ZS`) and youth unemployment,

ages 15-24 (`SL.UEM.1524.ZS`). Both are expressed as a percentage of the labour force. Coverage of the raw data is worldwide and annual, during 1991-2024.

AI use in enterprises - Eurostat
Out of all indicators provided in (`isoc_eb_ai`), we selected the percentage of enterprises with 10 or more employees that use at least one AI technology including speech recognition, machine learning, etc. (`E_AI_TANY`) The dataset covers mainly data of EU countries for the years 2021, 2023, 2024 and 2025, prompting the need to extract only the unemployment rate of youths and the total population in EU countries from the World Bank datasets.

Part 3 - Quantifying

3.1 Data cleaning

As we have chosen to focus on the years 2021-2024 for better comparison of the impact of AI before and after the launch of ChatGPT in 2022, we need to **filter data of unemployment within this period** (OpenAI, 2022).

Out of many dimensions available in the raw data source, we **filtered for number of enterprises using at least one of the AI technologies** (E_AI_TANY).

Table 3: Variables in the merged (combined) dataset

Variable	Type	Meaning
country	character	Country name
iso2c	character	2-letter country code
year	integer	Year of observation
total_unemp	numeric	Total unemployment (% of labour force)
youth_unemp	numeric	Youth unemployment, ages 15-24 (% of labour force)
values	numeric	Enterprises using at least one AI technology (%)
ai_group	factor	AI adoption group: Low / Medium / High (new variable)
youth_ratio	numeric	Youth-to-total unemployment ratio (new variable)

3.2 Generate necessary variables

Variable 1: AI adoption rate, or AI usage intensity

ai_group - bins the continuous values into Low, Medium, and High, so we can compare youth unemployment outcomes across different levels of AI adoption.

Variable 2:

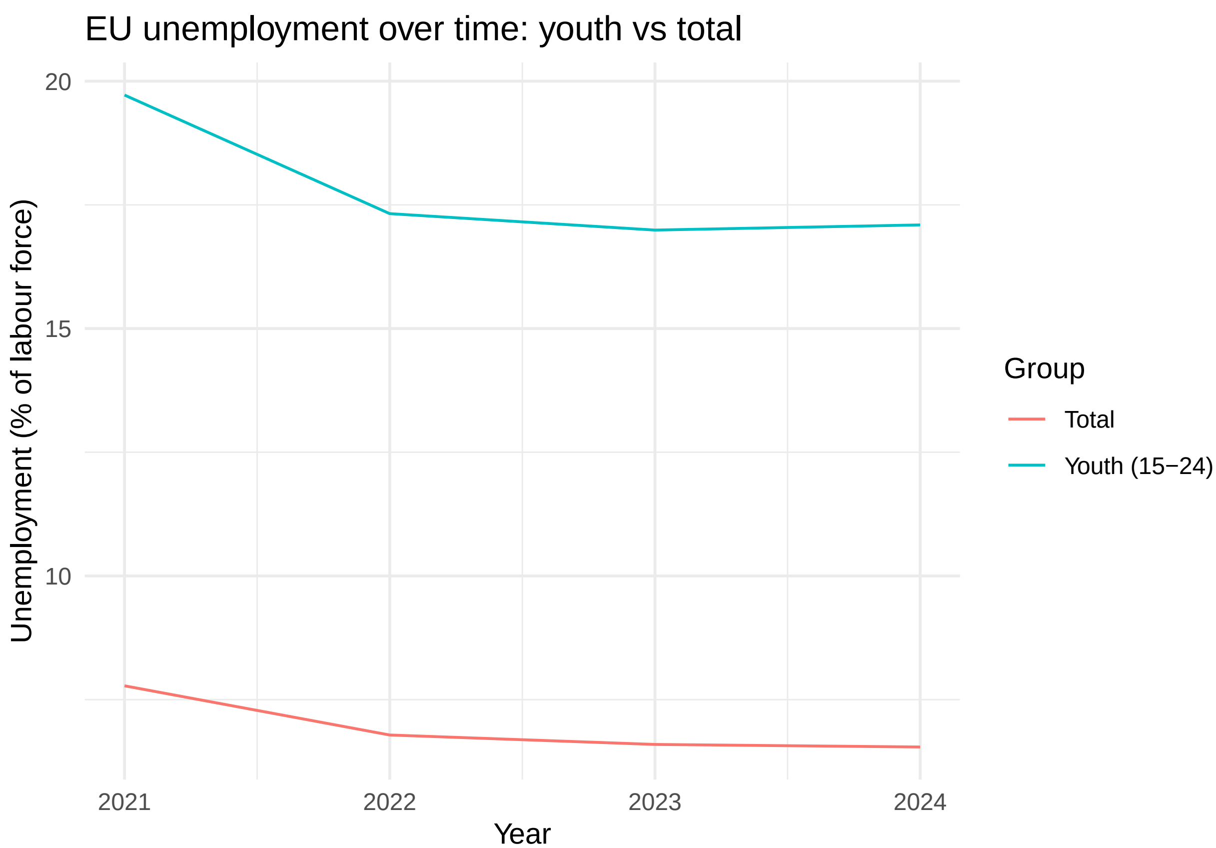
Youth_ratio is youth_unemp / total_unemp. It is the youth-to-total unemployment ratio. A value above 1 means young workers face higher unemployment than the workforce overall, which captures their relative disadvantage.

```
## # A tibble: 3 x 4
##   ai_group     n mean_total_unemp mean_youth_ratio
##   <chr>    <int>            <dbl>            <dbl>
## 1 High        41             7.76             2.59
## 2 Low         59             7.38             2.74
## 3 Medium      36             5.23             2.65
```

Table 4: Unemployment by level of AI adoption (2024)

ai_group	n	mean_total_unemp	mean_youth_unemp	mean_youth_ratio
High	2	11.14	25.52	2.28
Low	15	7.11	18.48	2.78
Medium	17	5.50	14.88	2.75

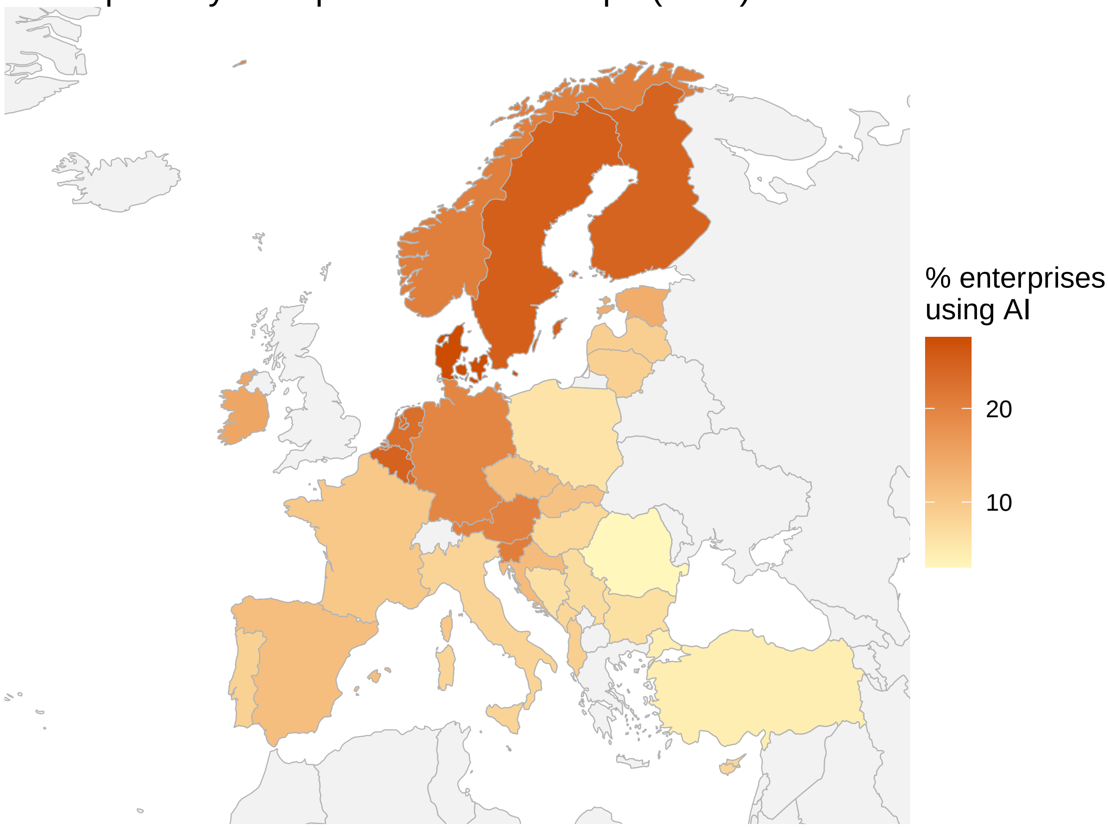
3.3 Visualize temporal variation



This plot tracks average youth (15–24 years old) and total unemployment across EU countries over time. Youth unemployment is consistently higher than the total rate in every year, and the two series move in similar manners, both rising in downturns and easing in recoveries, until 2023. In 2024, there is a slight difference as the total unemployment rate has been gradually decreasing since 2023, yet the youth unemployment rate keeps increasing since 2023. Knowing this is crucial for our question because it shows young workers are persistently the more vulnerable group. They are the segment that is most likely to be affected by the rise of AI usage in the labour market.

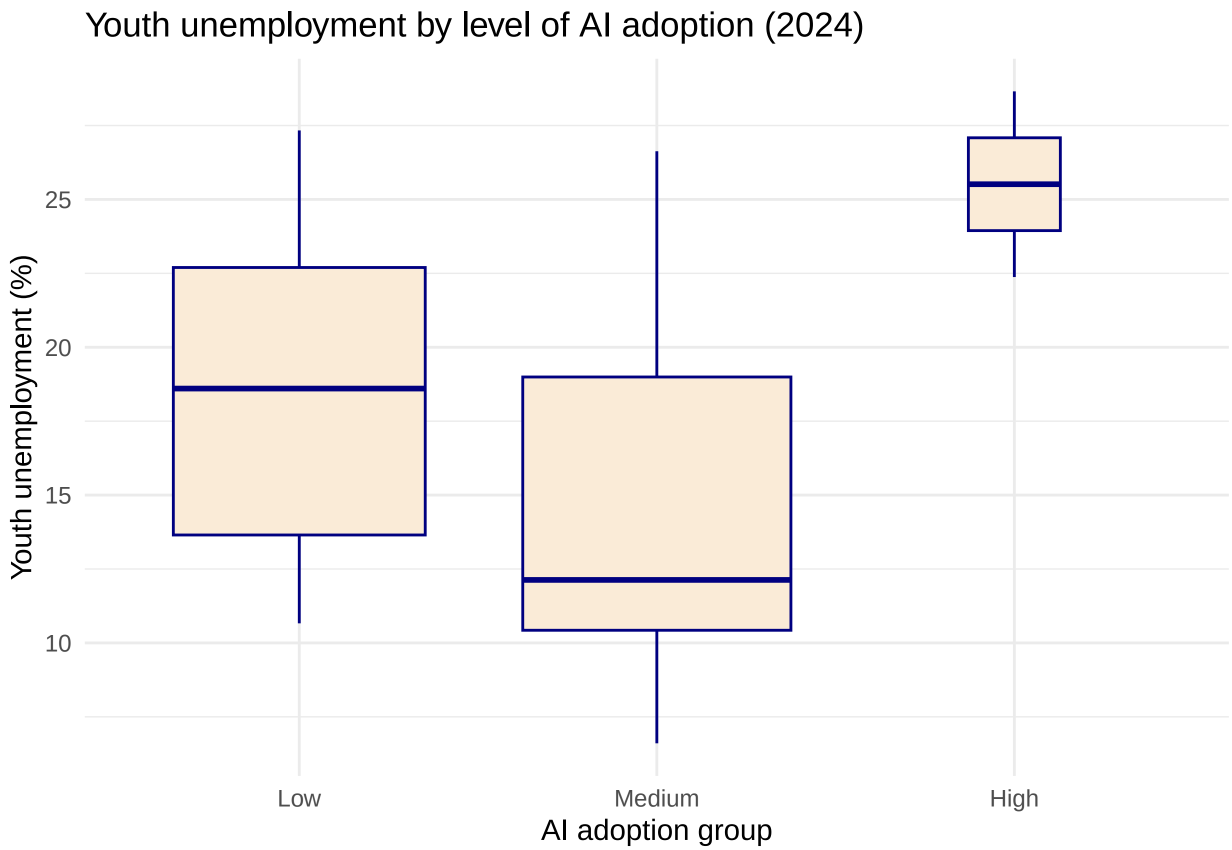
3.4 Visualize spatial variation

AI adoption by enterprises across Europe (2024)



The map shows each European country by the share of its enterprises using at least one AI technology in 2024. Adoption is uneven: it is highest in the Nordic countries (Denmark, Finland, Sweden) and parts of north-west Europe, and lowest across southern and eastern Europe. This geography is inmportant to our question because it identifies where AI is most embedded in the economy, and therefore where any AI-related effect on jobs should be most visible. A point worth taking notes is that countries with missing or insufficient data are shown in grey rather than left blank, so regions with the absence of colours signifies data gaps, not zero AI adoption.

3.5 Visualize sub-population variation

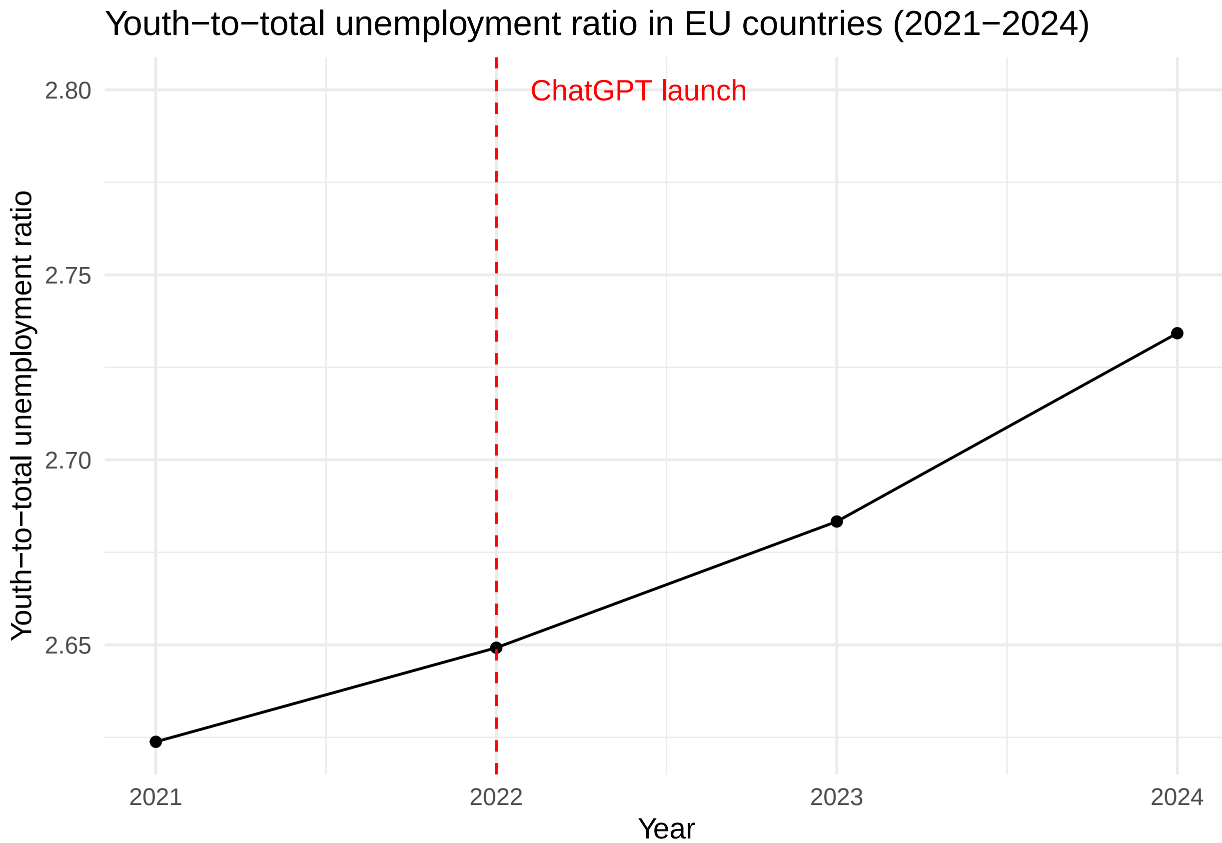


This boxplot compares youth unemployment across European countries grouped by AI-adoption level in 2024. Most countries fall into the Low (15 countries) or Medium (17 countries) groups; only two countries reach the High (>30%) group. Among the two well-populated groups, youth unemployment is lower in Medium-adoption countries (median around 12%) than in Low-adoption ones (around 18.5%), which is the opposite of the misconception of "more AI means fewer youth jobs" The boxes overlap substantially, so differences between groups are modest relative to the wide variation within each group. This suggests AI-adoption level alone does not explain youth unemployment: the two High-adoption countries happen to have high youth unemployment for reasons tied to their own labour markets, and low-adoption countries tend to be economies with structurally high youth unemployment (e.g. parts of southern Europe), rather than this being an effect of AI.

3.6 Event analysis

Analyze the relationship between two variables.

```
## # A tibble: 4 x 2
##    year mean_youth_ratio
##   <dbl>            <dbl>
## 1  2021             2.62
## 2  2022             2.65
## 3  2023             2.68
## 4  2024             2.73
```



The graph shows the average youth-to-total unemployment ratio across EU countries from 2021 to 2024. The ratio rises steadily after 2022 - the year ChatGPT launched - suggesting that young workers may be increasingly disadvantaged relative to the broader workforce as AI adoption grows. However, this is an association, not proof: other factors such as post-COVID labor market shifts or broader economic conditions could equally explain the trend.

Part 4 - Discussion

4.1 Discuss your findings

AI Adoption and Unemployment in European Countries

1. Youth unemployment is consistently much higher than overall unemployment
From 2021-2024, youth unemployment (ages 15-24) remained substantially above total unemployment in all years. The youth-to-total unemployment ratio ranged from 2.64 to 2.77, meaning young people were approximately 2.6-2.8 times more likely to be unemployed than the average worker.

2. The youth unemployment gap appears to be increasing
The youth-to-total unemployment ratio increased from 2.64 in 2021 to 2.77 in 2024. This suggests that young workers may have become increasingly disadvantaged relative to the broader workforce during the period in which AI adoption accelerated. However, this is an association and does not prove causation.

3. AI adoption across Europe is still low-to-moderate
In 2024 only two countries exceeded 30% enterprise adoption, so most European countries fall into the Low

or Medium groups. AI is not yet deeply embedded in the European economy.

4. Higher adoption is not associated with worse youth outcomes
Among the well-populated groups, Medium-adoption countries had lower average youth unemployment (14.9%) than Low-adoption ones (18.5%). The High group's higher unemployment rate (25.5%) rests on just two countries, so it reflects their specific labour markets rather than an AI effect. Overall there is no clear causal pattern between AI adoption and unemployment.

5. Youth unemployment remains disproportionately high regardless of AI level
Group differences are small next to within-group variation. The boxplots overlap heavily, and the youth-to-total ratio is similar across all groups (Low 2.78, Medium 2.75, High 2.28, indicating youth disadvantage is widespread across Europe regardless of AI level.

6. Country-specific differences appear important
The heatmap and boxplot suggest substantial variation between countries within each AI adoption category. This indicates that national labour market conditions, economic structures, and policy differences may play a larger role in shifting unemployment trends than AI adoption alone.

Limitations The relationship between AI adoption rate and unemployment is associational, not causal: the post-ChatGPT window is short, AI adoption measures enterprise use rather than job displacement, and confounders (the COVID recovery, differing economies) might play a larger role in explaining the variation. Therefore, we cannot conclude that the implementation of AI causes a worse-off unemployment trend for youths.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Cazzaniga, M., Jaumotte, F., Li, L., Melina, G., Panton, A. J., Pizzinelli, C., Rockall, E. J., Tavares, M. M., & Yao, K. (2024)

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Introducing chatgpt. (2022). OpenAI. Retrieved June 16, 2026, from https://openai.com/index/chatgpt/

World Bank. (2026). Unemployment, total (% of total labor force) (modeled ILO estimate) [Data set]. ILOSTAT. https://data.worldbank.org/indicator/SL.UEM.TOTL.ZS

World Bank. (2026). Unemployment, youth total (% of total labor force ages 15-24) (modeled ILO estimate) [Data set]. ILOSTAT. https://data.worldbank.org/indicator/SL.UEM.1524.ZS